Energy-Efficient Distributed Detection Through Feedback-Assisted Ordered Transmissions in the Presence of Fading and Quantization

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***Abstract*—We propose a novel energy-efficient feedback-enhanced successively reordered transmissions scheme (FE-SRTS) that combines distributed multiple access-based ordered channel access, feedback from the fusion node (FN), and quantized payloads. In FE-SRTS, the sensor nodes sequentially transmit their log-likelihood ratios (LLRs) to the FN until the latter decides. The order in which the nodes transmit is updated based on feedback from the FN and is implemented in a distributed manner using the timer scheme. We derive novel decision rules that enable FE-SRTS to achieve the detection error probability of the optimal rule in which the FN knows the LLRs of all nodes, but with a substantially lower average number of transmissions than conventional ordered and unordered schemes. We also account for retransmissions and power control due to fading, quantized payloads, and feedback. We analyze the average number of sensor transmissions and the total energy consumed by FE-SRTS. The analysis leads to insightful asymptotic results that establish the efficacy of FE-SRTS for Gaussian statistics. Our simulations, based on the Zigbee standard, show that the total energy consumed by FE-SRTS is markedly lower than by conventional schemes.**

***Index Terms*—Distributed detection, wireless sensor networks, ordered transmissions, feedback, energy-efficiency, quantization, fading channel.**

I. INTRODUCTION

DISTRIBUTED detection is a fundamental problem in signal detection theory impacting diverse fields such as surveillance, healthcare, environmental monitoring, transportation, and logistics [2]. Energy-efficiency is a crucial issue when distributed detection is implemented using a wireless sensor network (WSN), which consists of spatially separated sensor nodes and a fusion node (FN) or an edge server. While the FN may be connected to a power grid, the nodes are often powered by pre-charged batteries. The energies in the batteries drain out over time when the nodes measure a physical phenomenon, process it, and transmit some function of their measurements to the FN. Furthermore, large-scale deployments are being envisaged for WSNs [3]. Therefore, energy-efficiency improvement and network scalability are important themes of current research on WSNs [4], [5].

Several energy-efficient distributed schemes have been investigated in the literature. Clustering, duty cycling, and censoring curtail the number of times that a node transmits or the duration of transmission and reception to lower energy consumption [6], [7], [8], [9], [10], [11], [12], [13]. Clustering divides the nodes into clusters. The nodes in a cluster send their observations to the cluster head, which forwards the combined observation of the cluster to the FN [6], [7], [8], [9]. Duty cycling switches a node into sleep mode whenever it is not transmitting or receiving. Censoring curtails the number of transmitting sensors based on different parameters such as the false alarm probability [10], the communication rate [11], and the log-likelihood ratio (LLR) [12], [13].

Feedback from the FN has also been used to reduce energy consumption. The schemes in [14], [15], and [16] use feedback to iteratively update the local decisions at the nodes and the global decision at the FN. This approach requires fewer nodes to meet a target error probability. In contrast, the schemes in [17], [18], and [19] make the FN feed back quantized channel state information to the nodes to enable them to adapt their transmit powers. The finer the quantization, the lower the detection error probability. However, the payload size increases and so does the energy consumed. This trade-off has also been studied for censoring in [10] and [11] and for unordered transmissions scheme (UTS), in which all the nodes transmit their LLRs to the FN, in [17] and [18]. In [10], censoring and thresholds for one-bit quantization are derived to achieve near-optimal detection error probability. In [11], multi-bit quantization thresholds are derived.

However, the schemes in [6], [7], [8], [9], [10], [11], [12], [17], [18], [19], and [20] lead to a higher detection error probability, while those in [14], [15], and [16] make each node transmit at least once. The ordered transmissions scheme (OTS) is an exception to this norm. In OTS, fewer nodes transmit on average without any degradation in the detection error probability [21]. The nodes transmit in the decreasing order of the absolute value of their LLRs. When the FN receives an LLR from a node, it either decides a hypothesis

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and terminates all further transmissions or waits for one more transmission. OTS has been adopted for the Neyman-Pearson framework [22], correlated observations [23], slotted-Aloha access protocol [24], malicious nodes [25], [26], and energy harvesting [27]. It has been redesigned to exploit time diversity in fading channels in [13]. In [28], the decision thresholds of OTS are modified to reduce the average number of transmissions while achieving a near-optimal detection error probability.

A. Contributions

We propose a novel distributed detection scheme called feedback-enhanced successively reordered transmissions scheme (FE-SRTS) that combines: (1) distributed multiple access-based ordered channel access, (2) feedback from the FN, and (3) quantized payloads. In FE-SRTS, the sensor node with the maximum absolute value of LLR transmits its LLR to the FN. This is achieved in a distributed manner using the timer scheme, which does not require any node or the FN to know the LLRs of any other nodes in advance. The FN either decides, in which case it sends an early stop command to the nodes to stop their timers and no further transmissions occur, or it feeds back the received LLR to the nodes. In the next step, the node, with the largest absolute value of the difference between its LLR and the fed-back LLR, transmits.

Contrary to clustering, duty cycling, and censoring, FE-SRTS achieves the same optimal detection error probability as UTS [13], [14], [15], [16], [17], [18], [21], in which the FN knows the LLRs of all the nodes. An innovative aspect of FE-SRTS is the distributed reordering of the sequence in which the nodes transmit in each step. This is unlike OTS, in which the ordering is fixed. We make the following contributions.

- We derive a novel *alternating LLR ordering* lemma that establishes how feedback from the FN can be gainfully employed to make LLRs to the extreme left and right of the origin to be sent to the FN alternately. This leads to tighter bounds on the LLRs of the nodes yet to transmit to the FN. We then derive novel decision and feedback rules for the FN that achieve the same optimal detection error probability as UTS. These apply to arbitrary probability distributions for the measurements.
- We analyze the average number of transmissions of FE-SRTS for Gaussian statistics for the general case where both the mean and the variance of the hypotheses can be different. The shift-in-mean [21], [24], [25] and shift-in-variance [12], [27] problems are special cases of our model. We prove a remarkable property of FE-SRTS that for any number of nodes and any prior probability, its average number of transmitting sensors decreases to only two in the asymptotic regime of a large difference between the means or the ratio of variances of the two hypotheses.
- We address two important practical aspects in our model and analysis:
  - *Retransmissions,* which occur due to fading in the channel between the nodes and the FN, and noise. We adapt the transmit power of each node as a function of its channel power gain to achieve a target packet error rate (PER). The target PER can be lower for retransmissions. We show that the total energy consumed by the nodes and the FN in FE-SRTS is markedly lower than in UTS and OTS. While fading is considered in [13], error-free transmissions are assumed. Retransmissions and energy consumption have not been modeled in the ordered transmissions literatures [13], [21], [22], [23], [24], and [25].
  - *Quantized transmissions,* due to which the payloads transmitted by the nodes and FN must be from a pre-specified set. Here, we make three novel contributions. First, we present decision rules for FE-SRTS. These decision rules are based on *quantized information (QI)*, which can be interpreted to be the LLR of the quantized LLR. These rules are error-optimal unlike those in [17], [18], [19], and [22], which use quantized measurement-based decision rules. Second, we combine the proportional expansion approach of [29] with the timer scheme to avoid collision of packets due to multiple timers expiring simultaneously. To the best of our knowledge, this problem of quantization resulting in a strictly positive probability of collision has not been addressed in the ordered transmissions literatures [21], [22], [23], [24], and [25]. Third, we present a communication-efficient modification to the metric update and feedback generation rules. It requires only one bit of feedback in steps two and beyond and achieves the same optimal error probability as UTS.
- We benchmark the average number of transmissions and the total transmit energy consumed by the nodes and the FN. This benchmarking takes into account the modulation and overheads of the practical IEEE 802.15.4 Zigbee protocol. The total energy consumed by FE-SRTS is significantly lower than OTS and UTS, with the gap between them increasing with the number of nodes.

Table I highlights the key differences between FE-SRTS and several works on distributed detection.

B. Organization and Notations

Section II presents the system model. Section III specifies FE-SRTS, its decision rules, and analyzes its performance. Section IV considers fading. Section V specifies FE-SRTS with quantized LLRs. We present simulation results in Section VI, and our conclusions follow in Section VII.

Notations: The probability of an event A is denoted by $\Pr(A)$, and the conditional probability of A given event B by $\Pr(A|B)$. The expectation with respect to a random variable (RV) X is denoted by $\mathbb{E}_X[\cdot]$. We denote vectors in bold font. $\mathbb{1}_{[A]}$ denotes the indicator function; it equals 1 if A is true and is 0 otherwise. The notation $y \sim \mathcal{N}(\mu, \sigma^2)$ means that y is a Gaussian RV with mean μ and a variance σ^2, and $y \sim \mathcal{U}[a, b]$ means that y is an RV uniformly distributed over $[a, b]$. Transpose is denoted by $(\cdot)^T$.

TABLE I
KEY DIFFERENCES BETWEEN THE EXISTING WORKS AND FE-SRTS

Technique	*Works*	*Feedback*	*Quantization*	*Sensor node to FN channel*	*Error probability*
Clustering	[6], [8]	Yes	No	Noiseless	Sub-optimal
	[7]	Yes	No	Pathloss	Sub-optimal
Censoring	[10], [11]	No	Yes	Noiseless	Sub-optimal
	[12]	No	No	Noiseless	Sub-optimal
Feedback-assisted UTS	[17], [18], [19]	Yes	Yes	Fading	Sub-optimal
	[14], [15], [16]	Yes	No	Noiseless	Optimal
OTS	[21], [23], [24], [25], [26]	No	No	Noiseless	Optimal
	[28]	No	No	Noiseless	Sub-optimal
	[22]	No	Yes	Noiseless	Sub-optimal
Censored OTS	[13]	Yes	No	Fading but error-free	Optimal
Proposed (FE-SRTS)	-	Yes	Yes	Noiseless/Fading	Optimal

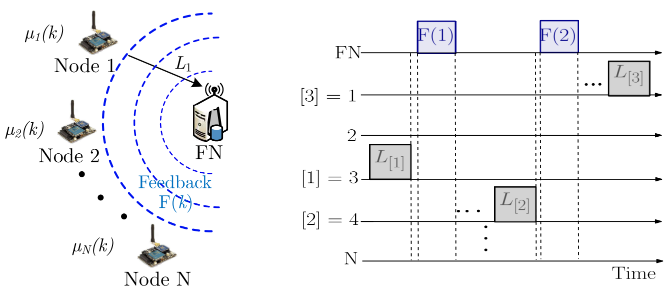


Fig. 1. System model showing measurements, LLRs, metrics of the nodes that determine which node transmits first, and the feedback by the FN.

II. SYSTEM MODEL

Consider a WSN with $N \geq 2$ geographically separated sensor nodes and an FN. Time is divided into measurement cycles. In each cycle, the FN has to decide based on measurements made by the nodes at the beginning of the cycle. We investigate binary hypotheses testing, which is a fundamental problem in signal detection theory [13], [17], [18], [21], [22], [23], [24]. The FN has to detect one of two hypotheses, namely a null hypothesis H_0 and a true hypothesis H_1. A hypothesis is a statement about population parameters such as mean, variance, and higher order moments [30, Ch. 8]. For example, in target detection, H_0 models the absence of a target, while H_1 models its presence. In spectrum sensing, H_1 models the presence of a primary signal, while H_0 models its absence. In healthcare, H_0 models the patient's blood pressure is less than the normal threshold and H_1 models an above-normal blood pressure.

Measurements, which are observations of the sensor nodes, are affected by the population parameter(s). For example, for target detection in a radar network, the received signal amplitude at the radar can be a measurement. In spectrum sensing, the received signal strength of a secondary user can be a measurement. The measurement y_i by node i has a conditional probability density function (PDF) of $f(y_i|H_h)$ under hypothesis H_h, for $h \in \{0,1\}$. According to the Bayesian detection framework, the optimum decision rule that minimizes the error probability is given by [31, Ch. II]

$$\sum_{i=1}^{N} L_i \underset{H_0}{\overset{H_1}{\gtrless}} \beta, \tag{1}$$

where $\beta = \log\left(\frac{(c_{10}-c_{00})\zeta_0}{(c_{01}-c_{11})\zeta_1}\right)$, c_{uv} is the cost incurred if hypothesis H_u is chosen while H_v is true, $\zeta_h > 0$ is the prior probability of H_h, and L_i is the LLR of node i. The LLR L_i is defined as

$$L_i = \log\left(\frac{f(y_i|H_1)}{f(y_i|H_0)}\right). \tag{2}$$

A conventional approach to implement the optimal rule in (1) is to make each node forward its LLR to the FN. The FN decides after receiving the LLRs from all the nodes. This requires N transmissions. We refer to this scheme as UTS.

III. FEEDBACK-ENHANCED SUCCESSIVELY REORDERED TRANSMISSIONS (FE-SRTS)

We now propose FE-SRTS that combines feedback from the FN and ordered transmissions to achieve significant reductions in the number of transmissions by the nodes and the energy consumed. We first discuss the unquantized case without fading to gain insights about the scheme. Sections IV and V focus on practical extensions that address fading and quantization.

FE-SRTS orders the transmissions of the nodes based on locally computed non-negative real numbers called *metrics*, which we specify below. The transmissions happen in a distributed manner using the timer scheme. Node i sets a timer that is a monotonically non-increasing function of the node's metric. Several functions for setting the timer have been discussed in the literatures [32] and [33]. Among them, a staircase-type function minimizes the probability that the FN cannot decode a packet because two or more timers expire at time instants close to each other [33].

When a node's timer expires, it transmits a packet containing its LLR to the FN. Thus, the node with the highest metric transmits first without having to know the metrics of the other nodes.[1] When the FN receives an LLR from a node, it uses the decision rules that we derive below to either decide or wait for more transmissions. If the FN decides, it sends an *early stop* command to terminate the measurement cycle and stop all further transmissions of the nodes. Otherwise, the FN feeds back the last received LLR to all the nodes. The nodes update their metrics upon receiving this feedback. Only a single transmission is needed for the early stop command or the feedback, as they can be broadcast by the FN.

Let $\mu_i(k)$ be the metric of node i at the beginning of the k^{th} step. The node with the largest metric in that step transmits.

[1] As in [21], [22], [23], [24], and [25], we assume that collisions due to timers expiring close to each other occur with negligible probability. This is because the probability that two continuous-valued metrics are the same is zero.

We denote its index as $[k]$. Let $\mathcal{S}_k$ denote the set of all nodes that have not transmitted at the beginning of the k^{th} step. $\mathcal{S}_1$ is initialized to $\{1, 2, \ldots, N\}$. The FN feeds back $F(k)$ if it does not decide after receiving $L_{[k]}$. Each step consists of the following four parts: metric update, ordered transmissions based on timer scheme, decision rule, and feedback. They differ for the first and subsequent steps.

- *Step* 1*:* Node i sets its metric as $\mu_i(1) = |L_i|$. Thus,
$$[1] = \arg\max_{i \in \mathcal{S}_1} \{\mu_i(1)\}, \tag{3}$$
and node $[1]$ is the first to transmit its LLR to the FN as its timer expires first. The FN either decides on a hypothesis and sends an early stop command; or it feeds back $F(1) = L_{[1]}$.
- *Step* 2*:* $\mathcal{S}_2 = \mathcal{S}_1 \setminus \{[1]\}$. Every node $i \in \mathcal{S}_2$ sets its metric as
$$\mu_i(2) = |L_i - F(1)|. \tag{4}$$
The nodes that remain in $\mathcal{S}_2$ run the timer scheme. Hence, node $[2]$ transmits in the second step, where
$$[2] = \arg\max_{i \in \mathcal{S}_2} \{\mu_i(2)\}. \tag{5}$$
The FN either decides and sends an early stop command or feeds back $F(2) = L_{[2]}$.
- *Step* $k \geq 2$*:* In general, $\mathcal{S}_k = \mathcal{S}_{k-1} \setminus \{[k-1]\}$. Node $i \in \mathcal{S}_k$ sets its metric as
$$\mu_i(k) = |L_i - F(k-1)| = \left|L_i - L_{[k-1]}\right|. \tag{6}$$
The nodes in $\mathcal{S}_k$ run the timer scheme. Hence, node $[k]$ transmits in step k where
$$[k] = \arg\max_{i \in \mathcal{S}_k} \{\mu_i(k)\}. \tag{7}$$
The FN either decides or it feeds back $F(k) = L_{[k]}$. If the FN decides, it sends an early stop command to all the nodes to stop their timers.
- The algorithm terminates once the FN has sent an early stop command or all the nodes have transmitted.

The feedback-assisted metric update rule in (6) and ordered transmissions lead to LLRs from the extreme left and right of the origin being alternatively sent to the FN. This is formally stated in the following lemma.

Lemma *1 (Alternating LLR Ordering):* For $2 \leq k \leq N-1$ and $k+1 \leq i \leq N$, the following statements hold:

- If $L_{[1]} \geq 0$ and k is even, or $L_{[1]} < 0$ and k is odd, then $L_{[k]} \leq L_{[i]} \leq L_{[k-1]}$.
- If $L_{[1]} \geq 0$ and k is odd, or $L_{[1]} < 0$ and k is even, then $L_{[k-1]} \leq L_{[i]} \leq L_{[k]}$.

Proof: The proof is given in Appendix A. ■

A. Decision Rules

The following decision rules at the FN achieve the same optimal error probability as UTS, where the FN knows all N LLRs.

Result *1:* For $k = 1$, the FN decides as follows:
$$\text{If } L_{[1]} \geq \beta + (N-1)\left|L_{[1]}\right|: \text{Decide } H_1, \tag{8}$$
$$\text{If } L_{[1]} < \beta - (N-1)\left|L_{[1]}\right|: \text{Decide } H_0, \tag{9}$$

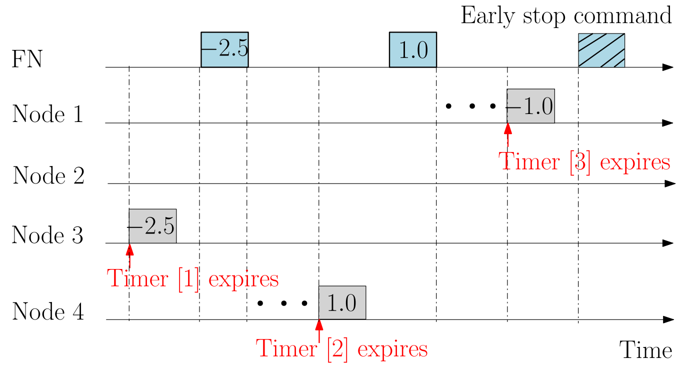


Fig. 2. Example of FE-SRTS with 4 nodes.

Else, feed back $F(1) = L_{[1]}$ and wait for next transmission. For $2 \leq k \leq N-1$, the FN decides as follows:

- If $L_{[1]} \geq 0$ and k is even, or $L_{[1]} < 0$ and k is odd, then
$$\text{If } \sum_{i=1}^{k} L_{[i]} \geq \beta - (N-k)L_{[k]}: \text{Decide } H_1, \tag{10}$$
$$\text{If } \sum_{i=1}^{k} L_{[i]} < \beta - (N-k)L_{[k-1]}: \text{Decide } H_0, \tag{11}$$
Else, feed back $F(k) = L_{[k]}$ and wait for the next transmission.
- If $L_{[1]} \geq 0$ and k is odd, or $L_{[1]} < 0$ and k is even, then
$$\text{If } \sum_{i=1}^{k} L_{[i]} \geq \beta - (N-k)L_{[k-1]}: \text{Decide } H_1, \tag{12}$$
$$\text{If } \sum_{i=1}^{k} L_{[i]} < \beta - (N-k)L_{[k]}: \text{Decide } H_0, \tag{13}$$
Else, feed back $F(k) = L_{[k]}$ and wait for the next transmission.

For $k = N$, the FN uses the rule in (1) to decide.

Proof: The proof is given in Appendix B. ■

For $k = 1$, the metric and the decision rule are the same as OTS. However, the metric and the decision rule are very different for $k \geq 2$ due to the above alternating LLR ordering. FE-SRTS terminates when the FN decides. In the worst case, the FN requires all N transmissions to decide. Thus, FE-SRTS is guaranteed to terminate. The duration of the transmissions is much smaller than the measurement cycle since the time steps in the timer scheme are much smaller than the measurement cycle duration.

1) Example: Consider a WSN with $N = 4$ and $\beta = 0$. Then, $\mathcal{S}_1 = \{1, 2, 3, 4\}$. Let the LLRs of the nodes be $L_1 = -1.0$, $L_2 = -0.5$, $L_3 = -2.5$, and $L_4 = 1.0$. Thus, $[1] = 3$. In this case, it turns out from (8) and (9) that the FN does not decide. Thus, it feeds back $F(1) = L_{[1]} = -2.5$. And, $\mathcal{S}_2$ is updated to $\{1, 2, 4\}$. The nodes in $\mathcal{S}_2$ set their metrics as per (4) to $\mu_1(2) = 1.5$, $\mu_2(2) = 2$, and $\mu_4(2) = 3.5$. Hence, $[2] = 4$. In this case, it turns out from (12) and (13) that the FN again does not make a decision and feeds back $F(2) = L_{[2]} = 1.0$. Then, $\mathcal{S}_3$ becomes $\{1, 2\}$. The nodes in

$\mathcal{S}_3$ set their metrics to $\mu_1(3) = 2.0$ and $\mu_2(3) = 1.5$. Thus, $[3] = 1$. In this case, it turns out that the FN decides in favor of H_0, and, thus, requires only three transmissions. On the other hand, UTS would have required four transmissions.

2) Extensions: FE-SRTS can be extended to multiple and composite hypotheses testing problems. In multiple hypotheses testing, the error-optimal decision rules are separable. For example, for the shift-in-mean testing problem, the decision rule simplifies to comparing the sum of the measurements with multiple thresholds [34, Ch. 3]. This separable form implies that FE-SRTS can be applied to it. The key idea behind designing the decision rule in each step of FE-SRTS is to find an upper bound and a lower bound on $\sum_{i=1}^{N} y_i$ given the ordered measurements that the FN has received thus far. The FN feeds back the measurements to reorder the nodes.

When the conditional distributions or their parameters are unknown, the detection problem can be viewed as a composite hypotheses testing problem. The generalized likelihood ratio test (GLRT) decision rule [34, Ch. 6] for it has a form similar to (1), except that a node computes its LLR using a maximum-likelihood estimate of the unknown parameters. For example, for the shift-in-mean problem, the maximum-likelihood estimate of unknown mean parameter of H_1 is the empirical mean of the measurements of all the nodes. It is shown in [34, Ch. 6] that the decision rule simplifies to comparing the sum of the measurements with two thresholds that depend on the prior probabilities. Thus, composite hypotheses testing can also be implemented using FE-SRTS.

B. Analysis: Average Number of Sensor Transmissions

We now derive an expression for the average number of transmitting sensors. Let $\boldsymbol{L} = (L_1, \ldots, L_N)$ and $\boldsymbol{l} = (l_1, \ldots, l_N)$ be a realization of $\boldsymbol{L}$. As specified by Lemma 1, $l_{[1]}, l_{[2]}, \ldots, l_{[N]}$ get transmitted to the FN. Let Φ_t be the region of LLRs $\boldsymbol{l}$ for which the FN decides in the t^{th} transmission. We characterize Φ_t first.

Lemma *2:* Φ_t is given as follows:

1) *For* $t = 1$:
$$\Phi_1 = \left\{\boldsymbol{l} : l_{[1]} - (N-1)\left|l_{[1]}\right| \geq \beta\right\} \cup \left\{\boldsymbol{l} : l_{[1]} + (N-1)\left|l_{[1]}\right| < \beta\right\}. \tag{14}$$

2) *For* $2 \leq t \leq N-1$:
- *For* $l_{[1]} \geq 0$ *and* t *is even, or* $l_{[1]} < 0$ *and* t *is odd:*
$$\Phi_t = \left\{\boldsymbol{l} : \sum_{i=1}^{t} l_{[i]} + (N-t)l_{[t]} \geq \beta\right\} \cup \left\{\boldsymbol{l} : \sum_{i=1}^{t} l_{[i]} + (N-t)l_{[t-1]} < \beta\right\}. \tag{15}$$
- *For* $l_{[1]} \geq 0$ *and* t *is odd, or* $l_{[1]} < 0$ *and* t *is even:*
$$\Phi_t = \left\{\boldsymbol{l} : \sum_{i=1}^{t} l_{[i]} + (N-t)l_{[t-1]} \geq \beta\right\} \cup \left\{\boldsymbol{l} : \sum_{i=1}^{t} l_{[i]} + (N-t)l_{[t]} < \beta\right\}. \tag{16}$$

3) *For* $t = N$, $\phi_N = \mathbb{R}^N$.

Proof: From the decision rules for $k = 1$ in (8) and (9), we see that the FN decides H_1 if $l_{[1]} - (N-1)\left|l_{[1]}\right| \geq \beta$ and it decides H_0 if $l_{[1]} + (N-1)\left|l_{[1]}\right| < \beta$. This yields (14). The logic is similar for $k \geq 2$ and is skipped to conserve space. ■

Using Lemma 2, we derive the average number of transmitting sensors $\bar{D}_{\text{tx}}(N)$.

Lemma *3:* $\bar{D}_{\text{tx}}(N)$ is given by
$$\bar{D}_{\text{tx}}(N) = \sum_{h \in \{0,1\}} \sum_{t=1}^{N} t \Pr\left(\boldsymbol{L} \in \Gamma_t | H_h\right) \zeta_h, \tag{17}$$
where $\Gamma_t = \Phi_1^c \cap \ldots \cap \Phi_{t-1}^c \cap \Phi_t$.

Proof: The proof follows from the law of total expectation. We skip the details to conserve space. ■

Comments: The probability in (17) can be numerically computed using Monte Carlo techniques. For a given t and H_h, we generate G i.i.d. realizations of $\boldsymbol{L}$, which we denote by $\boldsymbol{l}_1, \ldots, \boldsymbol{l}_G$. If a realization satisfies the condition $\boldsymbol{l}_n \in \Gamma_t$, then $\mathbb{1}_{[\boldsymbol{l}_n \in \Gamma_t]} = 1$, and is 0 otherwise. Its empirical mean $\frac{1}{G}\sum_{n=1}^{G} \mathbb{1}_{[\boldsymbol{l}_n \in \Gamma_t]}$ converges almost surely to $\Pr\left(\boldsymbol{L} \in \Gamma_t | H_h\right)$. The computational error decreases as $O\left(1/\sqrt{G}\right)$.

The above lemma has an involved form because the ordered LLRs $L_{[1]}, L_{[2]}, \ldots, L_{[N]}$ are not independent even though $L_1, L_2, \ldots, L_N$ are independent when conditioned on the hypothesis. The lemma provides an independent verification of the numerical simulation results.

It also leads to the following powerful asymptotic result that only two nodes are required to transmit for any N and ζ_1 for Gaussian statistics. Here, y_i is given by
$$y_i \sim \mathcal{N}\left(\mu_0, \sigma_0^2\right), \text{ under hypothesis } H_0, \tag{18}$$
$$y_i \sim \mathcal{N}\left(\mu_1, \sigma_1^2\right), \text{ under hypothesis } H_1, \tag{19}$$
where μ_h and σ_h^2 denote the mean and variance, respectively, under hypothesis H_h, for $h \in \{0, 1\}$. The shift-in-mean ($\sigma_0^2 = \sigma_1^2 < \infty$, $\mu_0 < \infty$) and shift-in-variance ($\mu_1 = \mu_0 < \infty$, $\sigma_0 < \infty$) problems come under our general formulation. We refer to σ_1^2/σ_0^2 as the HSNR and $\mu_1 - \mu_0$ as the mean-shift. These are important system parameters.

The Gaussian model is widely studied in the literatures [12], [13], [21], [24], [25], and [27]. It is justified when the sensor measurements are affected by electronic noise and fluctuations in environmental conditions. In such cases, the Gaussianity arises due to the central limit theorem. The measurements are conditionally independent when these noises and fluctuations are independent across nodes. The transmission rule and the decision rules of FE-SRTS apply to arbitrary probability distributions. We specify the conditional distributions as Gaussian to quantify the efficacy of FE-SRTS and to glean valuable and mathematically rigorous insights on its performance.

Result *2:* For any ζ_1 and N, the average number of transmitting sensors, $\bar{D}_{\text{tx}}(N)$, until the FN decides is
- *Shift-in-variance:* $\lim_{(\sigma_1/\sigma_0) \to \infty} \bar{D}_{\text{tx}}(N) = 2$.
- *Shift-in-mean:* $\lim_{(\mu_1 - \mu_0) \to \infty} \bar{D}_{\text{tx}}(N) = 2$.

Proof: The proof is given in Appendix C. ■

Result 2 applies regardless of the number of nodes and the prior probability. On the other hand, the average number

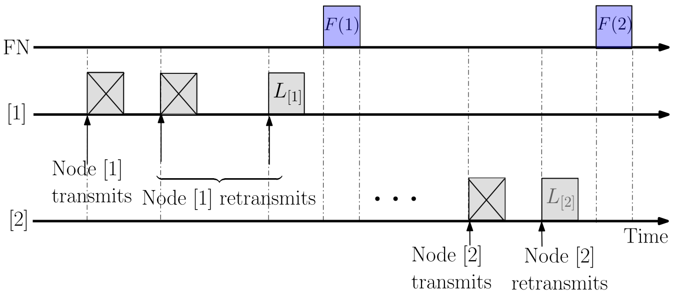


Fig. 3. FE-SRTS transmission rule in fading channels.

of transmitting sensors grows as $\lceil N/2 \rceil$ in OTS, where $\lceil \cdot \rceil$ denotes the ceiling function [21], and N in UTS.

IV. FE-SRTS Under Fading Channels

We now study the implications of fading in the channels between the nodes and the FN, and noise on the average number of transmissions and energy consumption. The probability that a packet cannot be decoded at the FN depends on the channel power gain and the transmit power of the node. If the FN cannot decode a packet containing a node's LLR, it cannot feed back that LLR. Thereafter, the updating of the metrics and the ordering of nodes are no longer possible. Therefore, retransmissions are necessary.

Let the power gain of the channel between node i and the FN be G_i. For Rayleigh fading, G_i is an exponential RV with mean δ_i, which is the pathloss from node i to the FN. The pathloss, in turn, depends on the distance between node i and the FN. The maximum distance between a node and the FN is $d_{\max}$. Let $\delta_{\max}$ be the pathloss at $d_{\max}$.

In a measurement cycle, when node i transmits with power P_i, its signal-to-noise ratio (SNR) γ_i at the FN is $P_i G_i/\sigma^2$, where σ^2 is the noise variance. For the modulation and coding scheme used by the node, let the minimum SNR required to achieve a PER of ε be $\gamma_{\text{th}}(\varepsilon)$. The FN cannot decode the packet from node i if $\gamma_i < \gamma_{\text{th}}(\varepsilon)$. This is justified given the waterfall nature of the PER as a function of the SNR.

A node sets its transmit power to meet the PER target. However, if the power required exceeds the peak transmit power, $P_{\max}$, the node does not participate in the scheme. If the FN decodes the node's transmission, then it transmits an early stop command or a feedback as before. Else, the FN does not transmit. In such a case, the transmitting node times out and retransmits. It does so until the FN sends back an early stop or a feedback.[2] To improve the odds of decoding the retransmitted packets, the nodes retransmit with a higher power that is sufficient to meet a tighter PER target.

A. Transmission Rule for FE-SRTS for Fading Channel

We highlight the key aspects where FE-SRTS differs from the no-fading case studied in Section III. We also specify the energy consumed in each step. It depends on the payloads of the packets sent by the nodes and the bit duration T_{bit}. Let B_{Nd}, B_{ES} and B_{FB} denote the payloads of the packet sent by the node, early stop command, and feedback, respectively.

At the start of a measurement cycle, node i computes its required transmit power $P_i = \gamma_{\text{th}}(\varepsilon_1)\,\sigma^2/G_i$ to transmit to the FN with a target PER of ε_1. If $P_i > P_{\max}$, node i does not participate in the measurement cycle. Let $\mathcal{S}_1(\boldsymbol{G}) \subseteq \{1, 2, \ldots, N\}$ denote the set of nodes that participate, where $\boldsymbol{G} = [G_1, \ldots, G_N]$.

Step 1: Participating node $i \in \mathcal{S}_1(\boldsymbol{G})$ sets its metric as $\mu_i(1) = |L_i|$. As before, $[k]$ denotes the node with the highest metric from $\mathcal{S}_1(\boldsymbol{G})$ in step k. As described in Section III, the timer of node [1] with the highest metric in $\mathcal{S}_1(\boldsymbol{G})$ expires first, where

$$[1] = \arg\max_{i \in \mathcal{S}_1(\boldsymbol{G})} \{\mu_i(1)\}. \tag{20}$$

Node [1] sends a packet to the FN containing $L_{[1]}$ with a transmit power $P_{[1]} = \gamma_{\text{th}}(\varepsilon_1)\,\sigma^2/G_{[1]}$. Hence, node [1] expends an energy of $\gamma_{\text{th}}(\varepsilon_1)\,\sigma^2 B_{\text{Nd}} T_{\text{bit}}/G_{[1]}$. The FN takes one of the following actions:

1) If it decodes the packet and cannot decide, it feeds back a packet containing $F(1) = L_{[1]}$ with power $\gamma_{\text{FB}}\sigma^2/\delta_{\max}$ to all nodes. The FN consumes an energy of $\gamma_{\text{FB}}\sigma^2 B_{\text{FB}} T_{\text{bit}}/\delta_{\max}$. Here, γ_{FB} is chosen such that a node at a distance $d_{\max}$ from the FN decodes a packet with a negligible PER.[3]
2) If it decodes the packet and decides using decision rules in (8) and (9), it sends an early stop command to all the nodes with a power of $\gamma_{\text{FB}}\sigma^2/\delta_{\max}$. The FN consumes an energy of $\gamma_{\text{FB}}\sigma^2 B_{\text{ES}} T_{\text{bit}}/\delta_{\max}$.
3) If it cannot decode the packet, it remains idle. This causes node [1] to time out and then retransmit its packet with a power $\min\{\gamma_{\text{th}}(\varepsilon_r)\,\sigma^2/G_{[1]}, P_{\max}\}$. Here, $\varepsilon_r \leq \varepsilon_1$ is the PER target for retransmissions. Thus, the energy consumed by the node per retransmission is $\min\{\gamma_{\text{th}}(\varepsilon_r)\,\sigma^2/G_{[1]}, P_{\max}\}\, B_{\text{Nd}} T_{\text{bit}}$. The node retransmits until it receives feedback or an early stop from the FN.[4]

Step $k \geq 2$: In general, $\mathcal{S}_k(\boldsymbol{G}) = \mathcal{S}_{k-1}(\boldsymbol{G}) \setminus \{[k-1]\}$. Node $i \in \mathcal{S}_k(\boldsymbol{G})$ sets its metric using (6). Among the nodes in $\mathcal{S}_k(\boldsymbol{G})$, node $[k] = \arg\max_{i \in \mathcal{S}_k(\boldsymbol{G})} \{\mu_i(k)\}$ ends up transmitting its LLR to the FN. The FN performs one of the three actions defined in Step 1. Based on the FN's action, the node $[k]$ does the following:

1) If it receives $F(k) = L_{[k]}$ or an early stop, it drops-off from the measurement cycle.
2) Else, it times out and retransmits with power $\min\{\gamma_{\text{th}}(\varepsilon_r)\,\sigma^2/G_{[k]}, P_{\max}\}$ until it receives an early stop or a feedback from the FN. The energy consumed per retransmission is $\min\{\gamma_{\text{th}}(\varepsilon_r)\,\sigma^2/G_{[k]}, P_{\max}\}\, B_{\text{Nd}} T_{\text{bit}}$.

[2] In practice, the number of retransmissions can be capped. Our analysis applies so long as the probability of the FN failing to decode the transmitted packet after the maximum number of retransmissions is negligible.

[3] The FN's PER target is set below the target detection error probability.

[4] We assume that a node successfully transmits its LLR to the FN before the timer of the next node expires. This is justified when the time-out duration is less than the duration between the expiry of two successive timers.

B. Analysis: Transmissions and Total Energy Consumed

The number of retransmissions ρ_i, for node i, is a discrete RV with probability mass function (PMF) given by

$$\Pr(\rho_i = j) = \begin{cases} 1-\varepsilon_1, & \text{for } j = 0, \\ \varepsilon_1 \left(\max\{\varepsilon_r, e_i\}\right)^{j-1}\left(1-\max\{\varepsilon_r, e_i\}\right), & \text{for } j \geq 1, \end{cases}$$

where e_i is the PER of node i if it transmits with power $P_{\max}$.

1) Average Number of Sensor Transmissions ($\bar{\Theta}_{tx}$): $\bar{\Theta}_{\text{tx}}$ differs from the average number of transmitting sensors $\bar{D}_{\text{tx}}(N)$, which is defined in (17), due to retransmissions. Let $\boldsymbol{\rho} = [\rho_1\ \rho_2\ \cdots\ \rho_N]^T$. We can show that

$$\bar{\Theta}_{\text{tx}} = \mathbb{E}_{\boldsymbol{G},\boldsymbol{\rho}}\left[\sum_{t=1}^{|\mathcal{S}_1(\boldsymbol{G})|}\left(t + \sum_{j=1}^{t}\rho_{[j]}\right)\Pr(\boldsymbol{L}\in\Gamma_t)\right]. \quad (21)$$

The above expression simplifies to the following insightful closed-form in the asymptotic power regime.

Result *3:* For $P_{\max}\to\infty$,

$$\lim_{P_{\max}\to\infty}\bar{\Theta}_{\text{tx}} = \bar{D}_{\text{tx}}(N)\left(1+\frac{\varepsilon_1}{1-\varepsilon_r}\right). \quad (22)$$

Proof: The proof is given in Appendix D. ■

Note that Result 3 holds in general for any ordered or unordered scheme even though we derived it for FE-SRTS. Combining Results 2 and 3 leads to the following corollary for Gaussian statistics that shows the efficacy of FE-SRTS.

Corollary *1:* For any ζ_1, the following results hold for FE-SRTS:

- *Shift-in-variance:* $\lim_{\substack{P_{\max}\to\infty \\ (\sigma_1/\sigma_0)\to\infty}}\bar{\Theta}_{\text{tx}} = 2+\frac{2\varepsilon_1}{1-\varepsilon_r}$.
- *Shift-in-mean:* $\lim_{\substack{P_{\max}\to\infty \\ (\mu_1-\mu_0)\to\infty}}\bar{\Theta}_{\text{tx}} = 2+\frac{2\varepsilon_1}{1-\varepsilon_r}$.

When the HSNR or the mean-shift is high, we can show that $\bar{\Theta}_{\text{tx}} = \lceil N/2\rceil(1+\varepsilon_1/(1-\varepsilon_r))$ for OTS and $\bar{\Theta}_{\text{tx}} = N(1+\varepsilon_1/(1-\varepsilon_r))$ for UTS. These are substantially higher than that of FE-SRTS.

When the FN decodes in each step, it sends a packet to prevent the transmitting node from further transmissions for OTS and UTS. For FE-SRTS, the feedback serves this purpose. Therefore, the average number of broadcasts by the FN depends on the average number of sensor transmissions, which, as we saw, is small for FE-SRTS.

2) Total Energy Consumed $\bar{E}_{tot}$ by Sensor Nodes and FN: We can show that $\bar{E}_{\text{tot}}$ is given by

$$\begin{aligned}\bar{E}_{\text{tot}} &= T_{\text{bit}}\mathbb{E}_{\boldsymbol{G},\boldsymbol{\rho}}\Bigg[B_{\text{Nd}}\sum_{t=1}^{|\mathcal{S}_1(\boldsymbol{G})|}\Bigg(\sum_{j=1}^{t}\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{G_{[j]}}\mathbb{1}_{\left[\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{G_{[j]}}\leq P_{\max}\right]} \\ &\quad + \sum_{j=1}^{t}\rho_{[j]}\min\left\{\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{G_{[j]}}, P_{\max}\right\}\Bigg)\Pr(\boldsymbol{L}\in\Gamma_t)\Bigg] \\ &\quad + \frac{T_{\text{bit}}\gamma_{\text{FB}}\sigma^2}{\delta_{\max}} \\ &\quad \times\left[B_{\text{ES}} + B_{\text{FB}}\left(\mathbb{E}_{\boldsymbol{G}}\left[\sum_{t=1}^{|\mathcal{S}_1(\boldsymbol{G})|}t\Pr(\boldsymbol{L}\in\Gamma_t)\right]-1\right)\right]. \end{aligned} \quad (23)$$

The first and second summands in the right hand side of (23) correspond to the total energy consumed by the nodes and the FN, respectively.

The above expression has an involved form because $\rho_{[j]}$, for $j\in\{1,2,\ldots,N\}$, depends on $\boldsymbol{G}$. It simplifies to the following closed-form when $\delta_1=\cdots=\delta_N=\delta$ and $e_1=\cdots=e_N\approx\varepsilon_r$:

$$\begin{aligned}\bar{E}_{\text{tot}} &\approx \frac{\bar{D}_{\text{tx}}(N)B_{\text{Nd}}T_{\text{bit}}\sigma^2}{\delta} \\ &\quad \Bigg[\gamma_{\text{th}}(\varepsilon_1)\zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{\delta P_{\max}}\right) + \frac{\varepsilon_1}{1-\varepsilon_r} \\ &\quad \times\left(\gamma_{\text{th}}(\varepsilon_r)\zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}\right)+\frac{\delta P_{\max}}{\sigma^2}\left(1-e^{-\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}}\right)\right)\Bigg] \\ &\quad + \frac{\gamma_{\text{FB}}\sigma^2T_{\text{bit}}}{\delta_{\max}}\left[B_{\text{ES}}+\left(\bar{D}_{\text{tx}}(N)-1\right)B_{\text{FB}}\right], \end{aligned} \quad (24)$$

where $\zeta(x)=\int_x^\infty e^{-t}/t\,dt$ denotes the exponential integral [35, Ch. 5]. The derivation of (24) is given in Appendix E.

Notice that $\bar{E}_{\text{tot}}$ is proportional to $\bar{D}_{\text{tx}}(N)$, which is markedly lower for FE-SRTS than UTS and OTS. For Gaussian statistics, Result 2 says that $\bar{D}_{\text{tx}}(N)=2$ for FE-SRTS when the HSNR or the mean-shift is large. Then,

$$\begin{aligned}\bar{E}_{\text{tot}} &\to \frac{2B_{\text{Nd}}T_{\text{bit}}\sigma^2}{\delta} \\ &\quad \Bigg[\gamma_{\text{th}}(\varepsilon_1)\zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{\delta P_{\max}}\right) + \frac{\varepsilon_1}{1-\varepsilon_r} \\ &\quad \times\left(\gamma_{\text{th}}(\varepsilon_r)\zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}\right)+\frac{\delta P_{\max}}{\sigma^2}\left(1-e^{-\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}}\right)\right)\Bigg] \\ &\quad + \frac{\gamma_{\text{FB}}\sigma^2T_{\text{bit}}}{\delta_{\max}}\left[B_{\text{ES}}+B_{\text{FB}}\right]. \end{aligned} \quad (25)$$

Thus, $\bar{E}_{\text{tot}}$ of FE-SRTS becomes independent of N. This is unlike OTS and UTS, where $\bar{E}_{\text{tot}}$ is proportional to N.

V. FE-SRTS With Quantization

We now consider the practical imperative that each node has to transmit a q-bit quantized version of a function of its LLR to the FN and not its analog value. Similarly, the feedback from the FN is also quantized. Here, q is a system parameter. For ease of exposition, we do not consider fading below. We do not analyze fading and quantization together as this makes the notation, model, and analysis difficult to follow and insights harder to glean.

Let $M = 2^q$ denote the total number of quantization thresholds. Let τ_m be the m^{th} threshold, with $-\infty\leq\tau_0<\tau_1<\cdots<\tau_M=\infty$. Thus, the quantized LLR $L_i^{(q)}$ of node i is given by

$$L_i^{(q)}=\tau_m \text{ if and only if } \tau_m\leq L_i<\tau_{m+1}. \quad (26)$$

We use the cumulative distribution function (CDF) quantizer [36]. It divides the CDF of the LLR uniformly into 2^q bins. Thus, the quantized values occur with equal probability. Let $F_L(.)$ be the CDF of the LLR and $F_L^{-1}(.)$ be its inverse function. Then,

$$\tau_m = F_L^{-1}(m\Delta), \text{ for } m\in\{0,1,\ldots,M\}, \quad (27)$$

where $\Delta = 2^{-q}$. Thus, $\Pr\left(L_i^{(q)} = \tau_j\right) = \Delta, \forall j \in \{0, 1, \ldots, M-1\}$. We note that the theory below applies to other quantizers as well.

With the quantized LLRs, the optimal decision rule that minimizes the error probability changes to [31, Ch. II]

$$\sum_{i=1}^{N} Z_i \underset{H_0}{\overset{H_1}{\gtrless}} \beta, \tag{28}$$

where $\beta = \log\left(\frac{(c_{10}-c_{00})\zeta_0}{(c_{01}-c_{11})\zeta_1}\right)$,

$$Z_i = \log\left(P\left(L_i^{(q)}|H_1\right) \Big/ P\left(L_i^{(q)}|H_0\right)\right), \tag{29}$$

and $P\left(L_i^{(q)}|H_h\right)$ is the PMF of $L_i^{(q)}$ conditioned on hypothesis h. In effect, Z_i is the LLR of $L_i^{(q)}$. We call Z_i the QI of node i. It takes 2^q values and can be represented using q bits. We present closed-form expressions for the conditional PMFs and QI for Gaussian statistics below.

Gaussian Statistics: The CDF $F_L(.)$ of the LLR can be shown to be

$$\begin{aligned} F_L(x) = \Bigg[&\zeta_0\left(Q\left(\frac{R_2(x)-\mu_0}{\sigma_0}\right) - Q\left(\frac{R_1(x)-\mu_0}{\sigma_0}\right)\right) \\ &+\zeta_1\left(Q\left(\frac{R_2(x)-\mu_1}{\sigma_1}\right) - Q\left(\frac{R_1(x)-\mu_1}{\sigma_1}\right)\right)\Bigg] \\ &\times \mathbb{1}_{\left[x \geq c - \frac{b^2}{4a}\right]}. \end{aligned} \tag{30}$$

where $Q(.)$ is the Gaussian Q-function [35, Ch. 26.2],

$$R_1(x) = \frac{-b + \sqrt{b^2 - 4a(c-x)}}{2a}, \tag{31}$$

$$R_2(x) = \frac{-b - \sqrt{b^2 - 4a(c-x)}}{2a}, \tag{32}$$

$a = \left(\sigma_1^2 - \sigma_0^2\right)/\left(2\sigma_0^2\sigma_1^2\right), b = \left(\mu_1/\sigma_1^2\right) - \left(\mu_0/\sigma_0^2\right)$, and $c = \log(\sigma_0/\sigma_1) + \left(\mu_0^2/2\sigma_0^2\right) - \left(\mu_1^2/2\sigma_1^2\right)$.

Lemma *4:* The conditional PMF of $L_i^{(q)}$ is given by

$$\begin{aligned} P\left(L_i^{(q)} = \tau_j|H_h\right) &= Q\left(\frac{R_1(\tau_j)-\mu_h}{\sigma_h}\right) \\ &- Q\left(\frac{R_1(\tau_{j+1})-\mu_h}{\sigma_h}\right) + Q\left(\frac{R_2(\tau_{j+1})-\mu_h}{\sigma_h}\right) \\ &- Q\left(\frac{R_2(\tau_j)-\mu_h}{\sigma_h}\right), \text{ for } h \in \{0,1\}. \end{aligned} \tag{33}$$

Proof: We skip the proof to conserve space. ■

Substituting (33) in (29) yields the expression for Z_i.

We now present the transmission rule and derive the decision rules for quantized FE-SRTS (QFE-SRTS). They achieve the same error probability as quantized UTS (Q-UTS) in which all N nodes transmit their QI to the FN.

A. QFE-SRTS Transmission and Decision Rules

As before, $[k]$ denotes the node that transmits its QI, $Z_{[k]}$, to the FN in step k. The transmission rule is as follows:

Step 1*:* $\mathcal{S}_1$ is set to $\{1, 2, \ldots, N\}$. Node i computes its QI, Z_i, using (29) and sets its metric as $\mu_i(1) = |Z_i|$. *A key difference compared to the unquantized case is that $\mu_i(1)$ is a discrete RV that takes values from the finite set $\Omega(1) = \{\omega_0(1), \omega_1(1), \ldots, \omega_{M-1}(1)\}$.* Specifically, $\mu_i(1) = \omega_j(1)$ when $L_i^{(q)} = \tau_j$. From the definition of QI in (29), we get

$$\omega_j(1) = \left|\log\left(\frac{P\left(L_i^{(q)} = \tau_j|H_1\right)}{P\left(L_i^{(q)} = \tau_j|H_0\right)}\right)\right|, \text{ for } 0 \leq j \leq M-1. \tag{34}$$

Note that $\omega_0(1), \omega_1(1), \ldots, \omega_{M-1}(1)$ are equally likely.

Unlike Section III, the probability that two nodes have the same metric is non-zero. This can result in the timers of different nodes expiring simultaneously. The resultant collision at the FN prevents it from decoding any of their packets. Let $\omega_0(1), \omega_1(1), \ldots, \omega_{M-1}(1)$ be sorted in ascending order as $\omega_{0:M-1}(1) \leq \omega_{1:M-1}(1) \leq \cdots \leq \omega_{M-1:M-1}(1)$, where $\omega_{m:M-1}(1)$ denotes the m^{th} smallest element of $\Omega(1)$.

We combine the *proportional expansion* approach of [29] with the timer scheme to address this problem. In it, each node i performs the following actions to transmit its packet:

- If $\mu_i(1) = \omega_{m:M-1}(1)$, then node i maps $\mu_i(1)$ to a new continuous-valued metric $V_i(1) \sim \mathcal{U}[m\Delta, (m+1)\Delta]$.
- Node i sets its timer as a monotonically non-increasing function of $V_i(1)$ as in the timer scheme.

The following properties of proportional expansion explain its efficacy. First, ordering is preserved: If $\mu_i(1) < \mu_j(1)$ then $V_i(1) < V_j(1)$. Second, the probability that $V_i(1) = V_j(1)$, for $i \neq j$, is 0 because $V_i(1)$ is a continuous RV. Thus, the probability that timers of two nodes expire close to each other is negligible. Third, if $\mu_i(1) = \mu_j(1)$, then it does not matter whether node i or j transmits first.

Node $[1]$, which has the largest QI, transmits the index of the value that $Z_{[1]}$ takes to the FN. If the FN decides, it sends an early stop command. Else, it feeds back $F(1) = Z_{[1]}$.

Step $k \geq 2$*:* In general, $\mathcal{S}_k = \mathcal{S}_{k-1} \setminus \{[k-1]\}$. Node $i \in \mathcal{S}_k$ sets its metric $\mu_i(k)$ as $|Z_i - F(k-1)| = \left|Z_i - Z_{[k-1]}\right|$. Hence, $\mu_i(k)$ takes values from the set $\Omega(k) = \{\omega_0(k), \omega_1(k), \ldots, \omega_{M-1}(k)\}$, where

$$\omega_j(k) = \left|\log\left(\frac{P\left(L_i^{(q)} = \tau_j|H_1\right)}{P\left(L_i^{(q)} = \tau_j|H_0\right)}\right) - Z_{[k-1]}\right|, \quad \text{for } 0 \leq j \leq M-1. \tag{35}$$

Each node i maps $\mu_i(k)$ into the continuous-valued metric $V_i(k)$ using proportional expansion. Node $[k] = \arg\max_{i \in \mathcal{S}_k}\{V_i(k)\}$ is the next to transmit $Z_{[k]}$ to the FN. If the FN decides, it sends an early stop command and the algorithm terminates. Otherwise, it feeds back $F(k) = Z_{[k]}$ to all the nodes. Therefore, the metric update and feedback generation rules for QFE-SRTS are:

- *Node $i \in \mathcal{S}_k$:* The node updates its metric as follows:

$$\mu_i(k) = \begin{cases} |Z_i|, & \text{for } k = 1, \\ |Z_i - F(k-1)|, & \text{for } k \geq 2. \end{cases} \tag{36}$$

- *FN:* If it does not decide after receiving $Z_{[k]}$, it feeds back $F(k) = Z_{[k]}$. Else, it sends an early stop command.

Similar to Lemma 1, the following property holds about the transmission order of the nodes in QFE-SRTS. The proof is similar to Appendix A.

Lemma *5:* For $2 \leq k \leq N-1$ and $k+1 \leq i \leq N$, the following statements hold:

- If $Z_{[1]} \geq 0$ and k is even, or $Z_{[1]} < 0$ and k is odd, then $Z_{[k]} \leq Z_{[i]} \leq Z_{[k-1]}$.
- If $Z_{[1]} \geq 0$ and k is odd, or $Z_{[1]} < 0$ and k is even, then $Z_{[k-1]} \leq Z_{[i]} \leq Z_{[k]}$. ■

Decision Rules: Lemma 5 leads to decision rules that have the same form as those in Result 1, except that L_i is replaced with Z_i. They achieve the same optimal error probability as Q-UTS. We do not repeat them here.

For Gaussian statistics, it can be shown that $Z_i \to L_i$ as $q \to \infty$ for $i \in \{1, 2, \ldots, N\}$. Thus, QFE-SRTS reduces to FE-SRTS for $q \to \infty$. Therefore, we interchangeably use FE-SRTS and QFE-SRTS ($q = \infty$).

B. Feedback Overhead Reduction

For $k \geq 2$, just one bit of feedback $F'(k) \in \{\pm 1\}$ from the FN is sufficient, as we show below. This reduces the feedback overhead and also the energy consumption. The following metric update and feedback generation rules achieve this:

- Node $i \in \mathcal{S}_k$ updates its metric as follows:

$$\mu_i'(k) = \begin{cases} |Z_i|, & \text{for } k = 1, \\ |Z_i - F(1)|, & \text{for } k = 2, \\ |Z_i - F(k-1)F(1)|, & \text{for } k \geq 3. \end{cases} \tag{37}$$

- If the FN does not decide after receiving $Z_{[k]}$, it feeds back $F'(k)$ as follows:

$$F'(k) = \begin{cases} Z_{[1]}, & \text{for } k = 1, \\ (-1)^{k-1}, & \text{for } k \geq 2. \end{cases} \tag{38}$$

The lemma below proves that the detection error probability does not change due to such one-bit feedback.

Lemma *6:* The metric update rule in (37) and the feedback generation rule in (38) achieve the same optimal detection error probability as Q-UTS.

Proof: The proof is given in Appendix F. ■

VI. Numerical Results

We now present Monte Carlo simulation results to compare the detection error probability, the average number of sensor transmissions, and the total energy consumed by the nodes and the FN of QFE-SRTS and several other schemes as a function of several system parameters such as the number of nodes, the quantization level, and the SNR. The simulations take into account both fading and quantization. We use a block fading model, in which the channel gains between the nodes and the FN remain constant for a measurement cycle. Furthermore, the nodes and the FN use the CDF quantizer, which is described in Section V. We average over 10^6 measurement cycles. We set $c_{uv} = 1$ if $u \neq v$, and $c_{uv} = 0$ if $u = v$. Unless specified, $\varepsilon_1 = 0.1$ and $\varepsilon_r = \varepsilon_1/10$.

A. Energy Consumption Model at Sensor Nodes

We use the IEEE 802.15.4-2020 Zigbee standard [37] to realistically model the various overheads and energy consumption. The carrier frequency is 2.45 GHz, the bandwidth is 5 MHz, and the data rate is 250 Kbps. A packet consists of 88 header and frame check bits in addition to the payload. Offset quadrature phase shift keying is the modulation scheme. The thermal noise variance is -136.98 dB, which corresponds to the noise figure of 10 dB at a room temperature of 300 K. The decoding threshold SNRs for different PERs are listed in Table II. These are obtained from an IEEE 802.15.4 physical layer simulation using MATLAB's communications toolbox.

TABLE II
Decoding Threshold the Target PER

Target PER (ε)	*Decoding threshold ($\gamma_{th}(\varepsilon)$)*
10^{-1}	-2.96 dB
10^{-2}	-1.60 dB
10^{-3}	-0.60 dB

The sensor nodes are deployed uniformly within a circular area of radius $d_{\max} = 100$ m. The FN is at the origin of the circle. The path loss δ_i of node i is given by $\delta_i = \alpha\,(d_i/d_0)^{-\eta}$, where d_0 is the break-point distance and η is the pathloss coefficient [38]. We set $d_0 = 10$ m, $\eta = 3.7$, and $\alpha = -68.2$ dB. The maximum transmit power $P_{\max}$ of a sensor node is -3 dBm. Hence, the fading-averaged maximum SNR $\max \alpha\,(d_{\max}/d_0)^{-\eta}/\sigma^2$ at a distance $d_{\max}$ is 0 dB.

B. Benchmarking

We compare QFE-SRTS with Q-UTS and the following schemes:

- *Q-OTS [21]:* The metric of a node is the absolute value of its QI. When a node's timer expires, it transmits a packet containing its QI to the FN until the latter decodes. The packet format and payload are the same as QFE-SRTS.
- *CE-OT [22]:* A sensor node transmits a packet containing only one-bit quantized measurement made by the node to reduce the communication overhead. The quantization threshold is based on $q = 1$ for the CDF quantizer. The nodes follow the same ordering as Q-OTS. The decision rule at the FN is designed accordingly.[5]

Comparisons of the average number of sensor transmissions or the total energy consumed by censoring, duty cycling, and clustering are not shown as their error probabilities are larger. We present results for the shift-in-mean and shift-in-variance problems.

C. Shift-in-Mean Problem

Fig. 4 plots the detection error probability as a function of the mean-shift $\mu_1 - \mu_0$ for QFE-SRTS for $q = 1$, 2, and 8. Since the detection error probability of QFE-SRTS is the same as that of Q-UTS for any q, we do not plot Q-UTS. As q increases, the error probability decreases. This is because more fine-grained information is transmitted by the nodes and fed back by the FN. As ε_1 decreases, the error probability increases for any q. This is because the average number of

[5]While CE-OT is originally designed for the Neyman-Pearson framework, we adapt it for the Bayesian framework that we use. We optimize its decision threshold to minimize the detection error probability.

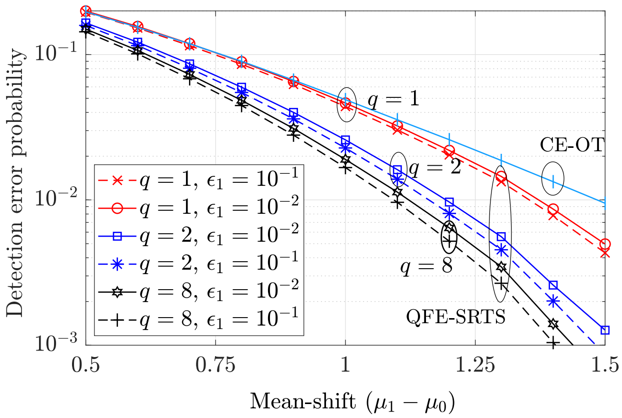


Fig. 4. Shift-in-mean: Detection error probability of QFE-SRTS as a function of the mean-shift ($\zeta_1 = 0.5$, $\mu_0 = 0$, and $\sigma_0 = \sigma_1 = 1$).

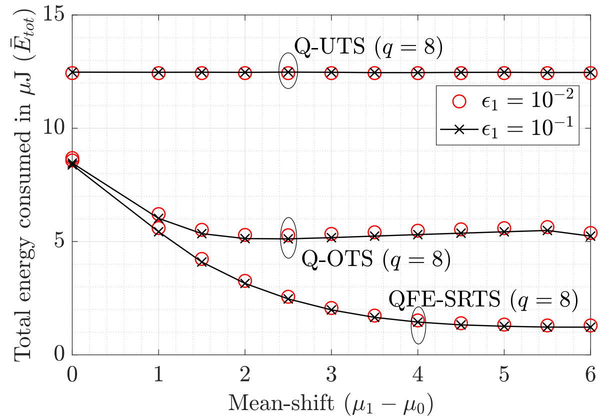
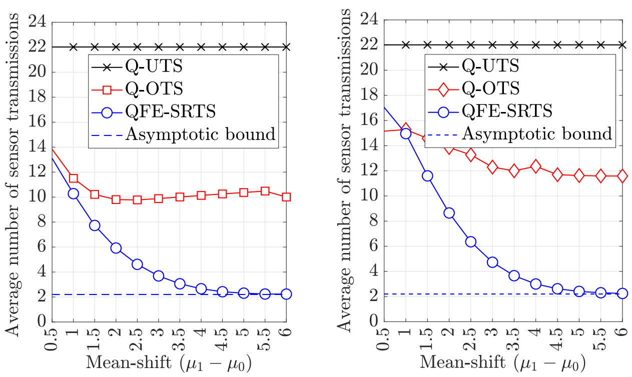


Fig. 6. Shift-in-mean: Total energy consumed as a function of the mean-shift ($\zeta_1 = 0.5$, $\mu_0 = 0$, and $\sigma_0 = \sigma_1 = 1$).



(a) $q = 8$ bits

(b) $q = 1$ bit

Fig. 5. Shift-in-mean: Average number of sensor transmissions of QFE-SRTS, Q-UTS, and Q-OTS as a function of the mean-shift ($\zeta_1 = 0.5$, $\mu_0 = 0$, and $\sigma_0 = \sigma_1 = 1$).

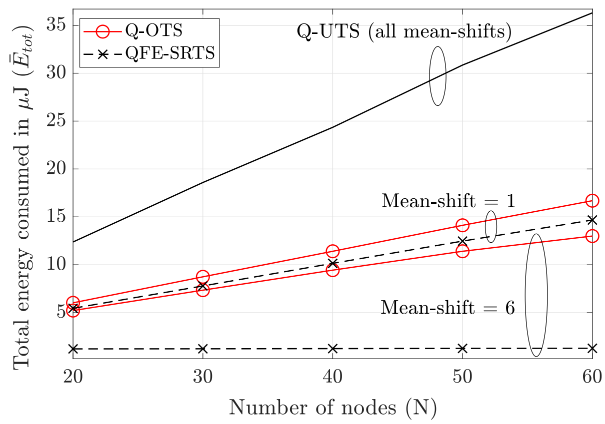


Fig. 7. Shift-in-mean: Total energy consumed as a function of N ($\zeta_1 = 0.5$, $\mu_0 = 0$, $\sigma_0 = \sigma_1 = 1$, and $q = 8$).

nodes participating in a measurement cycle decreases since $\gamma_{\text{th}}(\varepsilon_1)$ increases. As the mean-shift increases, the error probability decreases because the separation between the conditional distributions in (18) and (19) increases. Fig. 4 also plots the detection error probability of CE-OT. It is larger than that of QFE-SRTS. Hence, the quantized measurement-based decision rule, which CE-OT employs, is sub-optimal compared to the QI-based decision rule.

Fig. 5(a) compares the average number of sensor transmissions $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS, Q-UTS, and Q-OTS as a function of the mean-shift for $q = 8$. As the mean-shift increases, $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS decreases and saturates to 2.2 once $\mu_1 - \mu_0$ exceeds 5. Thus, the asymptotic regime for the mean-shift in Corollary 1 applies when $\mu_1 - \mu_0 \geq 5$. At $\mu_1 - \mu_0 = 2$, the average number of sensor transmissions of QFE-SRTS is 73.2% lower than that of Q-UTS and 39.7% lower than that of Q-OTS. At $\mu_1 - \mu_0 = 6$, the corresponding reductions are 90.0% and 78.0%, respectively.

Fig. 5(b) presents the corresponding result for $q = 1$. Even with $q = 1$, QFE-SRTS requires far fewer transmissions than Q-UTS. It also requires fewer transmissions than Q-OTS except when $\mu_1 - \mu_0$ is small. However, this regime is not of practical interest because the detection error probability is close to 0.5.[6] The asymptotic regime for the mean-shift in Corollary 1 applies when $\mu_1 - \mu_0 \geq 5$.

Fig. 6 compares the total energy consumed by the different schemes as a function of the mean-shift. $\bar{E}_{\text{tot}}$ decreases as the mean-shift increases because $\bar{\Theta}_{\text{tx}}$ decreases. As ε_1 decreases from 0.1 to 0.01, $\bar{E}_{\text{tot}}$ increases, albeit marginally. This is because $\gamma_{\text{th}}(\varepsilon_1)$ and $\gamma_{\text{th}}(\varepsilon_r)$ increase, which causes the node to transmit with higher power and consume more energy. At $\mu_1 - \mu_0 = 2$, QFE-SRTS consumes 74.7%, and 38.6% less energy than Q-UTS and Q-OTS, respectively. At $\mu_1 - \mu_0 = 6$, the corresponding numbers are 90.1% and 76.6%.

Fig. 7 plots the total energy consumed by the different schemes as a function of N for two values of the mean-shift. At a mean-shift of 1, QFE-SRTS consumes 56.3% to 59.6% lower energy than Q-UTS and 10.5% to 12.1% lower energy than Q-OTS. When the mean-shift increases to 6, QFE-SRTS consumes 90.1% to 96.5% lower energy than Q-UTS and 76.6% to 90.2% lower energy than Q-OTS. This is because, at the high mean-shift, the energy consumed by QFE-SRTS is

[6]In this regime, the one-bit quantized QI takes the values z and $-z$, where z is a small number. Thus, in Q-OTS, any node is selected with equal probability in each step. In step k, the sum QI at the FN lies in $[-kz, kz]$. However, QFE-SRTS alternatively selects the nodes with the QI z and $-z$. Thus, the sum QI at the FN lies in $[0, z]$ or $[0, -z]$ depending on the polarity of $Z_{[1]}$ until all the nodes with QI of either z or $-z$ have transmitted. Therefore, Q-OTS is more likely to cross one of its decision thresholds faster than QFE-SRTS.

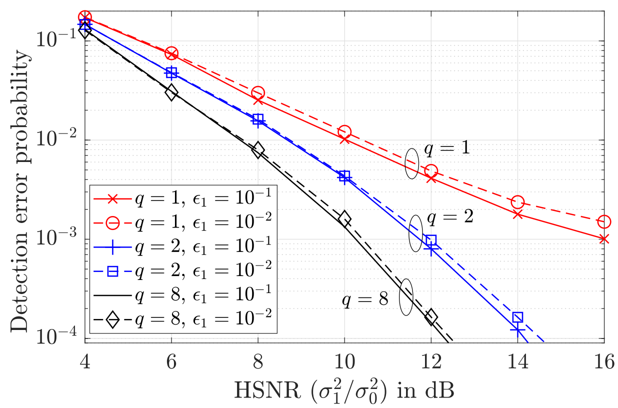


Fig. 8. Shift-in-variance: Detection error probability for QFE-SRTS as a function of the HSNR ($\zeta_1 = 0.5$, $\mu_0 = \mu_1 = 0$, and $\sigma_0 = 1$).

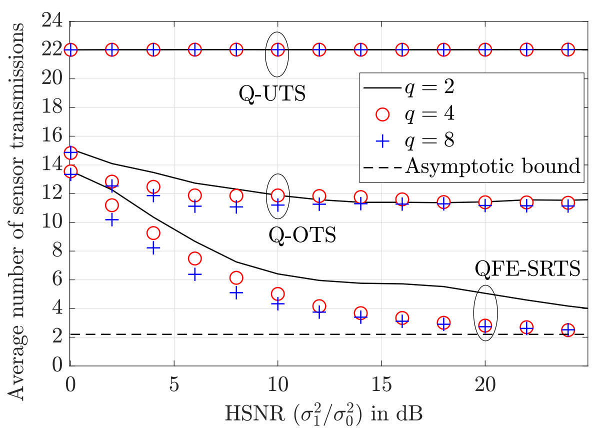


Fig. 9. Shift-in-variance: Average number of sensor transmissions as a function of the HSNR for different q ($\zeta_1 = 0.5$, $\mu_0 = \mu_1 = 0$, $\sigma_0 = 1$).

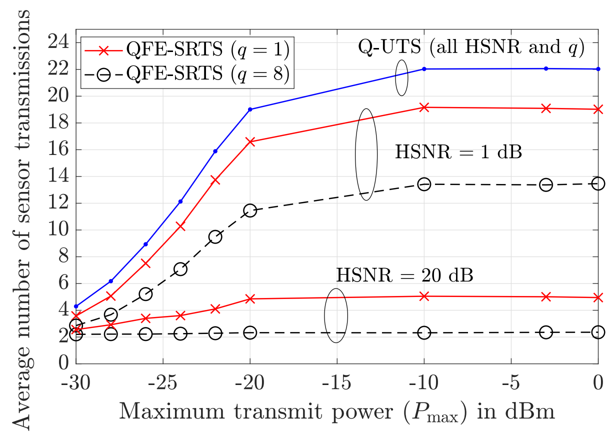


Fig. 10. Shift-in-variance: Average number of sensor transmissions as a function of $P_{\max}$ ($\zeta_1 = 0.5$, $\mu_0 = \mu_1 = 0$, and $\sigma_0 = 1$).

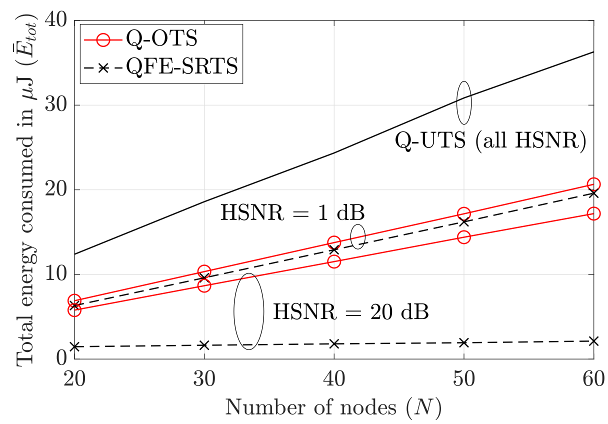


Fig. 11. Shift-in-variance: Total energy consumed as a function of N ($\zeta_1 = 0.5$, $\mu_0 = \mu_1 = 0$, $\sigma_0 = 1$, and $q = 8$).

constant while the energies consumed by Q-OTS and Q-UTS increase with N.

D. Shift-in-Variance Problem

Fig. 8 plots the detection error probability as a function of the mean-shift for QFE-SRTS for $q = 1$, 2, and 8. The detection error probability decreases as the HSNR increases. This is because the separation between the conditional distributions in (18) and (19) increases. The trends are similar to those for the shift-in-mean problem. As q increases, the detection error probability decreases because more fine-grained information is used for detection. The trend is more dominant at higher HSNRs. For example, at HSNR $= 12$ dB, the error probability of $q = 8$ is one order of magnitude less than that of $q = 1$. As ε_1 decreases, the detection error probability increases because fewer nodes participate on average.

Fig. 9 compares $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS, Q-UTS, and Q-OTS as a function of the HSNR for different quantization levels. As the HSNR increases, $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS decreases and saturates to 2.2. For $q = 8$, the asymptotic regime of the HSNR, at which $\bar{\Theta}_{\text{tx}}$ saturates, is HSNR ≥ 20 dB. For Q-UTS, $\bar{\Theta}_{\text{tx}}$ is always 22.0. This is 10.2% more than the number of nodes due to retransmissions. For any q and HSNR, QFE-SRTS requires markedly fewer transmissions, on average, than both Q-OTS and Q-UTS. For example, at HSNR $= 10$ dB and $q = 2$, $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS is 76.1% lower than that of Q-UTS and 54.9% lower than that of Q-OTS. At HSNR $= 20$ dB and $q = 2$, the corresponding reductions are 87.3% and 75.5%.At HSNR $=$ 10 dB and $q = 4$, the average number of sensor transmissions of QFE-SRTS is 80.5% lower than that of Q-UTS and 61.5% lower than that of Q-OTS. At HSNR $= 20$ dB and $q = 4$, the corresponding numbers are 87.5% and 76.1%.

Fig. 10 plots $\bar{\Theta}_{\text{tx}}$ of QFE-SRTS and Q-UTS as a function of $P_{\max}$ for $q = 1$ and 8 bits and for two values of the HSNR. We skip Q-OTS to avoid clutter. Since the nodes are at different distances from the FN, their pathlosses and SNRs (for the same transmit power) are different. By varying $P_{\max}$, we vary the SNRs of all the nodes. We observe that as $P_{\max}$ increases, $\bar{\Theta}_{\text{tx}}$ increases. This is because more nodes participate on average in a measurement cycle. $\bar{\Theta}_{\text{tx}}$ saturates at $P_{\max} = -10$ dBm. Here, all the nodes participate with high probability in a measurement cycle and the asymptotic regime of Result 3 now applies. Notice that at the high HSNR of 20 dB and $q = 8$, the average number of sensor transmissions is only 2.2.

Fig. 11 plots the total energy consumed by QFE-SRTS, Q-UTS, and Q-OTS as a function of N for two values of the HSNR. At an HSNR of 1 dB, QFE-SRTS consumes 48.6% to 53.3% lower energy than Q-UTS and 5.0% to 7.4% lower energy than Q-OTS. When the HSNR increases to 20 dB, QFE-SRTS consumes 88.3% to 95.6% lower energy than Q-UTS and 74.1% to 87.1% lower energy than Q-OTS. This is because, at high HSNR, the energy consumed by QFE-SRTS is constant while the energies consumed by Q-OTS and Q-UTS increase with N.

VII. Conclusion

We proposed a novel distributed detection scheme called FE-SRTS in which the metrics of the nodes, which determined the order in which they transmitted, were updated in each step based on the feedback from the FN. The feedback enabled LLRs from the extreme left and right of the origin to be alternately sent to the FN in a distributed manner. This led to novel decision rules that guaranteed the same optimal detection error probability as the conventional UTS but without the FN requiring LLRs from all the nodes. We extended FE-SRTS to account for power control and retransmissions due to fading. For quantized payload and feedback, the decision rules were based on QI. We combined the proportional expansion approach with the timer scheme to address the problem of timers expiring simultaneously due to the discrete metrics. Just one bit of feedback was sufficient to order the nodes from the second step onwards. We saw that in FE-SRTS, the average number of transmitting sensors decreased to just two for any prior probability and for any number of nodes in the asymptotic regime of a large HSNR or mean-shift. Consequently, the total energy consumed was much lower than conventional ordered and unordered schemes.

We focused on the detection problem in which the measurements are independent across nodes. In practice, the measurements can be correlated. This leads to a new challenge that the optimal Bayesian detection rule is no longer in a separable form that is amenable to a distributed implementation. Incorporating feedback to order the transmissions in this setting is an interesting avenue for future work. Another interesting direction is to extend FE-SRTS in the scenario where the nodes have imperfect knowledge of the channel gains. This affects the transmit powers of the nodes. This, in turn, affects the PER and the number of sensor transmissions. Lastly, an extension of FE-SRTS to a system with multiple FNs is of practical interest. One example of this is clustering. Here, the nodes can communicate in an energy-efficient manner with the their cluster heads using FE-SRTS. Furthermore, the cluster heads can communicate in an energy-efficient manner with the FN, again using FE-SRTS.

Appendix

A. Proof of Lemma 1 Using Mathematical Induction

We consider the three cases: (1) $L_{[1]} \geq 0$ and k even, (2) $L_{[1]} \geq 0$ and k odd, (3) $L_{[1]} < 0$ and k even, and (4) $L_{[1]} < 0$ and k odd, separately.

1. *$L_{[1]} \geq 0$ and k even:*

- *Initial Case ($k = 2$):* In step 1, since $\mu_i(1) = |L_i|$, for $1 \leq i \leq N$, and $[1]$ is the node with the highest metric, it follows from (3) that $\left|L_{[i]}\right| \leq \left|L_{[1]}\right| = L_{[1]}$. Therefore,

$$L_{[i]} \leq L_{[1]}, \text{ for } 2 \leq i \leq N. \tag{39}$$

In the step 2, the metric of node i is $\mu_i(2) = \left|L_i - F(1)\right| = \left|L_i - L_{[1]}\right|$. By the definition in (4) of $[2]$, we have $\left|L_{[2]} - L_{[1]}\right| \geq \left|L_{[i]} - L_{[1]}\right|$, for $3 \leq i \leq N$. This implies

$$L_{[1]} - \left|L_{[2]} - L_{[1]}\right| \leq L_{[i]} \leq L_{[1]} + \left|L_{[2]} - L_{[1]}\right|. \tag{40}$$

From (39), we know that $L_{[2]} \leq L_{[1]}$. Therefore, $\left|L_{[2]} - L_{[1]}\right| = L_{[1]} - L_{[2]}$. Substituting this in the upper bound in (40) yields $L_{[2]} \leq L_{[i]}$, for $3 \leq i \leq N$. Combining this with (39) yields $L_{[2]} \leq L_{[i]} \leq L_{[1]}$, for $3 \leq i \leq N$.

- *General Case:* Let Lemma 1 be true up to $k-1$. Since k is even, we are given that

$$L_{[k-2]} \leq L_{[i]} \leq L_{[k-1]}, \text{ for } k \leq i \leq N. \tag{41}$$

In the k^{th} step, the metric of node i is $\mu_i(k) = \left|L_i - F(k-1)\right| = \left|L_i - L_{[k-1]}\right|$. From the definition in (7) of $[k]$, we have $\left|L_{[k]} - L_{[k-1]}\right| \geq \left|L_{[i]} - L_{[k-1]}\right|$, for $k+1 \leq i \leq N$. It follows that

$$L_{[k-1]} - \left|L_{[k]} - L_{[k-1]}\right| \leq L_{[i]}, \text{ for } k+1 \leq i \leq N. \tag{42}$$

From (41), we know that $L_{[k]} \leq L_{[k-1]}$. Hence, $\left|L_{[k]} - L_{[k-1]}\right| = L_{[k-1]} - L_{[k]}$. Substituting this in (42), we get $L_{[k]} \leq L_{[i]}$, for $k+1 \leq i \leq N$. Combining this with (41) yields $L_{[k]} \leq L_{[i]} \leq L_{[k-1]}$, for $k+1 \leq i \leq N$.

2. *When k is Odd and $L_{[1]} \geq 0$:* Here, the initial case is $k = 3$. The proof follows a similar logic and is skipped.

3. *When $L_{[1]} < 0$:* The proofs for k even and k odd follow a similar logic and are skipped.

B. Decision Rules for FE-SRTS

The technique we use is similar to [21], which is to bound $\sum_{i=1}^{N} L_i$ based on the received LLRs. However, a crucial new ingredient in our approach is Lemma 1. It leads to tighter bounds on $\sum_{i=1}^{N} L_i$. We consider the three cases $k = 1$, $2 \leq k \leq N-1$, and $k = N$ separately below.

1) *When $k = 1$:* Since $L_{[1]}, L_{[2]}, \ldots, L_{[N]}$ is a permutation of $L_1, \ldots, L_N$, the decision rule in (1) can be recast as

$$\sum_{i=1}^{N} L_{[i]} \underset{H_0}{\overset{H_1}{\gtrless}} \beta. \tag{43}$$

Since $\left|L_{[i]}\right| \leq \left|L_{[1]}\right|$, for $2 \leq i \leq N$, we get $-\left|L_{[1]}\right| \leq L_{[i]} \leq \left|L_{[1]}\right|$. This implies that $-(N-1)\left|L_{[1]}\right| \leq \sum_{i=2}^{N} L_{[i]} \leq (N-1)\left|L_{[1]}\right|$. Hence,

$$L_{[1]} - (N-1)\left|L_{[1]}\right| \leq \sum_{i=1}^{N} L_{[i]} \leq L_{[1]} + (N-1)\left|L_{[1]}\right|. \tag{44}$$

Thus, if $L_{[1]} + (N-1)\left|L_{[1]}\right| < \beta$, it follows that $\sum_{i=1}^{N} L_{[i]} < \beta$ and the FN can decide H_0. Similarly, $L_{[1]} - (N-1)\left|L_{[1]}\right| \geq \beta$ implies that $\sum_{i=1}^{N} L_{[i]} \geq \beta$. Hence, the FN can decide H_1. This yields (8) and (9).

2) *When $2 \leq k \leq N$:* We consider the following two cases:

a) *$L_{[1]} \geq 0$ and k is Even, or $L_{[1]} < 0$ and k is Odd:* From Lemma 1, we have

$$L_{[k]} \leq L_{[i]} \leq L_{[k-1]}, \text{ for } k+1 \leq i \leq N. \quad (45)$$

Therefore, $(N-k)L_{[k]} \leq \sum_{i=k+1}^{N} L_{[i]} \leq (N-k)L_{[k-1]}$. Hence, $\sum_{i=1}^{k} L_{[i]} + (N-k)L_{[k]} \leq \sum_{i=1}^{N} L_{[i]} \leq \sum_{i=1}^{k} L_{[i]} + (N-k)L_{[k-1]}$.
If $\sum_{i=1}^{k} L_{[i]} + (N-k)L_{[k-1]} < \beta$, then $\sum_{i=1}^{N} L_{[i]} < \beta$. Hence, the FN can decide H_0. Similarly, if $\sum_{i=1}^{k} L_{[i]} + (N-k)L_{[k]} \geq \beta$, then $\sum_{i=1}^{N} L_{[i]} \geq \beta$ and the FN can decide H_1. This yields (10) and (11).

b) *$L_{[1]} \geq 0$ and k is Odd, or $L_{[1]} < 0$ and k is Even:* The logic is similar and is skipped.

3) *When $k = N$:* The FN knows all N LLRs. Hence, the optimal decision rule in (1) applies.

C. Proof of Result 2

1) Shift-in-Variance: We focus on the $t = 2$ term in (17). We can show that $\Pr(D_{\text{tx}} = 2|H_h) = \Pr(\boldsymbol{L} \in \Gamma_2|H_h)$. From Lemma 3, we get $\Gamma_2 = \Phi_1^c \cap \Phi_2$. Hence,

$$\begin{aligned}\Pr(D_{\text{tx}} = 2|H_h) &= \Pr(\boldsymbol{L} \in \Phi_1^c \cap \Phi_2|H_h), \\ &\geq \Pr(\boldsymbol{L} \in \Phi_1^c \cap \Phi_2, L_{[1]} \geq 0|H_h). \end{aligned} \quad (46)$$

The evaluation of the above probability term depends on whether β is positive or negative.

a) When $\beta \geq 0$: From (14), it follows that

$$\begin{aligned}\Phi_1^c &= \{\boldsymbol{l} : l_{[1]} - (N-1)\left|l_{[1]}\right| < \beta\} \\ &\cap \{\boldsymbol{l} : l_{[1]} + (N-1)\left|l_{[1]}\right| \geq \beta\}. \end{aligned} \quad (47)$$

For $l_{[1]} \geq 0$, (47) reduces to $\Phi_1^c = \{\boldsymbol{l} : l_{[1]} > -\beta/(N-2)\} \cap \{\boldsymbol{l} : l_{[1]} \geq \beta/N\}$. Hence,

$$\{\boldsymbol{l} \in \Phi_1^c, l_{[1]} \geq 0\} = \left\{\boldsymbol{l} : l_{[1]} \geq \frac{\beta}{N}, l_{[1]} \geq 0\right\}. \quad (48)$$

Next, we evaluate $\{\boldsymbol{l} \in \Phi_2, l_{[1]} \geq 0\}$. From (15), for $l_{[1]} \geq 0$, it follows that

$$\begin{aligned}\Phi_2 &= \{\boldsymbol{l} : (N-1)l_{[1]} + l_{[2]} < \beta\} \\ &\cup \{\boldsymbol{l} : l_{[1]} + (N-1)l_{[2]} \geq \beta\}. \end{aligned} \quad (49)$$

Hence, $\Phi_2 \supseteq \{\boldsymbol{l} : l_{[1]} + (N-1)l_{[2]} \geq \beta\}$. Therefore,

$$\{\boldsymbol{l} \in \Phi_2, l_{[1]} \geq 0\} \supseteq \{\boldsymbol{l} : l_{[1]} + (N-1)l_{[2]} \geq \beta, l_{[1]} \geq 0\}.$$

From Lemma 1, we know that $l_{[1]} \geq l_{[2]}$ if $l_{[1]} \geq 0$. Since $l_{[2]} \geq \beta/N$, it follows that $l_{[1]} + (N-1)l_{[2]} \geq \beta$. Therefore,

$$\{\boldsymbol{l} \in \Phi_2, l_{[1]} \geq 0\} \supseteq \left\{\boldsymbol{l} : l_{[2]} \geq \frac{\beta}{N}, l_{[1]} \geq 0\right\}. \quad (50)$$

Combining (48) and (50), we get

$$\{\boldsymbol{l} \in \Phi_1^c \cap \Phi_2, l_{[1]} \geq 0\} \supseteq \left\{\boldsymbol{l} : l_{[1]} \geq \frac{\beta}{N}, l_{[2]} \geq \frac{\beta}{N}, l_{[1]} \geq 0\right\}.$$

Furthermore, $l_{[1]} \geq \beta/N, l_{[2]} \geq \beta/N, \ldots, l_{[N]} \geq \beta/N$ implies $l_{[1]} \geq \beta/N$, $l_{[2]} \geq \beta/N$, and $l_{[1]} \geq 0$. Hence,

$$\{\boldsymbol{l} \in \Phi_1^c \cap \Phi_2, l_{[1]} \geq 0\} \supseteq \bigcup_{i=1}^{N} \left\{l_{[i]} \geq \frac{\beta}{N}\right\}. \quad (51)$$

Therefore, we can lower bound the following probability as:

$$\begin{aligned}&\Pr(\boldsymbol{L} \in \Phi_1^c \cap \Phi_2, L_{[1]} \geq 0|H_h) \\ &\geq \Pr\left(\bigcup_{i=1}^{N}\left\{L_{[i]} \geq \frac{\beta}{N}\right\} \middle| H_h\right) = \prod_{i=1}^{N} \Pr\left(L_i \geq \frac{\beta}{N} \middle| H_h\right), \end{aligned} \quad (52)$$

where the last step follows because the LLRs are independent when conditioned on H_h.

From (18) and (19), the LLR of node i in (2) is given by

$$L_i = a y_i^2 + b y_i + c, \quad (53)$$

where $a = (\sigma_1^2 - \sigma_0^2)/(2\sigma_0^2\sigma_1^2), b = (\mu_1/\sigma_1^2) - (\mu_0/\sigma_0^2)$, and $c = \log(\sigma_0/\sigma_1) + (\mu_0^2/2\sigma_0^2) - (\mu_1^2/2\sigma_1^2)$.

From (31), (32), and (53), $L_i \geq \beta/N$ if and only if $y_i \geq R_1(\beta/N)$ or $y_i \leq R_2(\beta/N)$. Hence,

$$\Pr\left(L_i \geq \frac{\beta}{N} \middle| H_1\right) = T_1 + T_2, \quad (54)$$

where $T_1 = Q((R_1(\beta/N) - \mu_1)/\sigma_1)$ and $T_2 = Q((R_2(\beta/N) - \mu_1)/\sigma_1)$.

For $\mu_1 = \mu_0$ and $\sigma_0 < \infty$, as $(\sigma_1/\sigma_0) \to \infty$, we can show from (31) that $R_1(\beta/N) \to \sigma_0\sqrt{2\log(\sigma_1/\sigma_0)}$. Therefore,

$$T_1 \to \lim_{(\sigma_1/\sigma_0)\to\infty} Q\left(\frac{\sigma_0}{\sigma_1}\sqrt{2\log\frac{\sigma_1}{\sigma_0}}\right) = Q(0) = 0.5. \quad (55)$$

Similarly, we can show that $\lim_{(\sigma_1/\sigma_0)\to\infty} T_2 = 0.5$.

Substituting this and (55) in (54), we get $\lim_{(\sigma_1/\sigma_0)\to\infty} \Pr(L_i \geq (\beta/N) \mid H_1) = 1$. Hence, from (52),

$$\lim_{(\sigma_1/\sigma_0)\to\infty} \Pr(\boldsymbol{L} \in \Phi_1^c \cap \Phi_2, L_{[1]} \geq 0|H_1) \geq 1. \quad (56)$$

This implies that $\lim_{(\sigma_1/\sigma_0)\to\infty} \Pr(\boldsymbol{L} \in \Phi_1^c \cap \Phi_2, L_{[1]} \geq 0|H_1) = 1$ since a probability cannot exceed one. Therefore, $\lim_{(\sigma_1/\sigma_0)\to\infty} \Pr(D_{\text{tx}} = 2|H_1) = 1$. Similarly, we can show that $\lim_{(\sigma_1/\sigma_0)\to\infty} \Pr(D_{\text{tx}} = 2|H_0) = 1$.

From the law of total probability, we know that

$$\Pr(D_{\text{tx}} = 2) = \sum_{h\in\{0,1\}} \zeta_h \Pr(D_{\text{tx}} = 2|H_h). \quad (57)$$

Substituting the above limits, we get $\Pr(D_{\text{tx}} = 2) = 1$.

b) When $\beta < 0$: The logic is similar and is skipped.

2) Shift-in-Mean: The logic is similar and is skipped.

D. Proof of Result 3

For $P_{\max} \to \infty$, $\mathcal{S}_1(\boldsymbol{G}) = \{1, 2, \ldots, N\}$ and the RV $\rho_{[j]}$ is independent of $\boldsymbol{G}$. Therefore, from (21), we get

$$\begin{aligned}\lim_{P_{\max}\to\infty} \bar{\Theta}_{\text{tx}} &= \sum_{t=1}^{N} t\Pr(\boldsymbol{L} \in \Gamma_t) \\ &+ \lim_{P_{\max}\to\infty} \sum_{t=1}^{N}\sum_{j=1}^{t} \mathbb{E}\left[\rho_{[j]}\right] \Pr(\boldsymbol{L} \in \Gamma_t). \end{aligned} \quad (58)$$

For $P_{\max} \to \infty$, we know that $e_{[j]} \to 0$. Hence, $\max\{\varepsilon_r, e_{[j]}\} = \varepsilon_r$, for all $j \in \mathcal{S}_1(\boldsymbol{G})$. Hence, $\Pr(\rho_{[j]} = 1) = 1 - \varepsilon_1$ and $\Pr(\rho_{[j]} = k) = \varepsilon_1\varepsilon_r^{k-1}(1 - \varepsilon_r)$, for $k \geq 1$ and $1 \leq j \leq N$.

From the law of total expectation, we have

$$\lim_{P_{\max}\to\infty} \mathbb{E}\left[\rho_{[j]}\right] = \sum_{i=0}^{\infty} i \lim_{P_{\max}\to\infty} \Pr\left(\rho_{[j]} = i\right), \tag{59}$$

$$= \varepsilon_1 \left(1-\varepsilon_r\right) \sum_{i=1}^{\infty} i\varepsilon_r^{i-1} = \frac{\varepsilon_1}{1-\varepsilon_r}. \tag{60}$$

Substituting (60) in (58), we get $\lim_{P_{\max}\to\infty} \bar{\Theta}_{\text{tx}} = (1+\varepsilon_1/(1-\varepsilon_r)) \sum_{t=1}^{N} t\Pr\left(\boldsymbol{L}\in\Gamma_t\right)$. Substituting (17) in the above expression yields (22).

E. Proof of (24)

We use the upper bound $|\mathcal{S}_1(\boldsymbol{G})| \leq N$. Furthermore, the RV $\rho_{[j]}$ is independent of $G_{[j]}$ because $e_{[j]} \approx \varepsilon_r$. Therefore, from (23), we get

$$\begin{aligned}
&\bar{E}_{\text{tot}} \\
&\approx T_{\text{bit}} B_{\text{Nd}} \sum_{t=1}^{N} \left(\sum_{j=1}^{t} \mathbb{E}_{\boldsymbol{G}} \left[\frac{\gamma_{\text{th}}\left(\varepsilon_1\right)\sigma^2}{G_{[j]}} \mathbb{1}_{\left[\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{G_{[j]}} \leq P_{\max}\right]} \right] \right. \\
&\left. + \sum_{j=1}^{t} \mathbb{E}\left[\rho_{[j]}\right] \mathbb{E}_{\boldsymbol{G}} \left[\min\left\{ \frac{\gamma_{\text{th}}\left(\varepsilon_r\right)\sigma^2}{G_{[j]}}, P_{\max} \right\} \right] \right) \Pr\left(\boldsymbol{L}\in\Gamma_t\right) \\
&+ \frac{T_{\text{bit}}\gamma_{\text{FB}}\sigma^2}{\delta_{\max}} \left[B_{\text{ES}} + B_{\text{FB}} \left(\sum_{t=1}^{N} t\Pr\left(\boldsymbol{L}\in\Gamma_t\right) - 1 \right) \right].
\end{aligned} \tag{61}$$

$G_{[j]}$ is an exponential RV with mean δ, for $j \in \{1,\ldots,N\}$ and $G_{[1]},\ldots,G_{[N]}$ are independent since the ordering is based on metrics, which are independent of $\boldsymbol{G}$. Hence,

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{G}} \left[\frac{\gamma_{\text{th}}\left(\varepsilon_1\right)\sigma^2}{G_{[j]}} \mathbb{1}_{\left[\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{G_{[j]}} \leq P_{\max}\right]} \right] \\
&= \int_{\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{P_{\max}}}^{\infty} \frac{\gamma_{\text{th}}\left(\varepsilon_1\right)\sigma^2 e^{-\frac{g}{\delta}}}{\delta g}\, dg = \frac{\gamma_{\text{th}}\left(\varepsilon_1\right)\sigma^2 \zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_1)\sigma^2}{\delta P_{\max}}\right)}{\delta},
\end{aligned} \tag{62}$$

where $\zeta(.)$ is the exponential integral. Similarly, we get

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{G}} \left[\min\left\{ \frac{\gamma_{\text{th}}\left(\varepsilon_r\right)\sigma^2}{G_{[j]}}, P_{\max} \right\} \right] = \frac{\gamma_{\text{th}}\left(\varepsilon_r\right)\sigma^2 \zeta\left(\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}\right)}{\delta} \\
&+ P_{\max}\left(1 - e^{-\frac{\gamma_{\text{th}}(\varepsilon_r)\sigma^2}{\delta P_{\max}}}\right).
\end{aligned} \tag{63}$$

The RVs $\rho_{[1]},\ldots,\rho_{[N]}$ are i.i.d. with $\Pr\left(\rho_{[j]}=1\right) = 1-\varepsilon_1$ and $\Pr\left(\rho_{[j]}=k\right) = \varepsilon_1\varepsilon_r^{k-1}\left(1-\varepsilon_r\right)$, for $k \geq 1$ and $1 \leq j \leq N$. Hence, $\mathbb{E}\left[\rho_{[j]}\right] = \varepsilon_1/(1-\varepsilon_r)$, for $j \in \{1,\ldots,N\}$.

Substituting the above expression along with (62), (63), and (17) in (61) yields (24).

F. Proof of Lemma 6

We shall show that the metric update rule in (37) and the feedback generation rule in (38) achieve the same sequence of sensor node transmissions as with q-bit feedback from the FN. This proves that the detection error probability is same as that of QFE-SRTS, which we know has the same optimal detection error probability as Q-UTS.

- *Step* 1*:* From (37) and (36), we get $\mu_i'(1) = \mu_i(1)$. Hence, $[1] = \arg\max_{i\in\mathcal{S}_1} \{\mu_i'(1)\} = \arg\max_{i\in\mathcal{S}_1} \{\mu_i(1)\}$.
- *Step* 2*:* From (38), we get $F'(1) = Z_{[1]} = F(1)$. Combining this with (37) and (36), we get $\mu_i'(2) = \mu_i(2)$. Hence, $[2] = \arg\max_{i\in\mathcal{S}_2} \{\mu_i'(2)\} = \arg\max_{i\in\mathcal{S}_2} \{\mu_i(2)\}$.
- *Step* 3*:* After step $k = 2$, the FN feeds back $F'(2) = -1$. In step $k = 3$, node i sets its metric as $\mu_i'(3) = \left|Z_i - F(2)Z_{[1]}\right|$. As a result, $[3] = \arg\max_{i\in\mathcal{S}_3} \left\{\left|Z_i + Z_{[1]}\right|\right\}$, which is the same as $\arg\max_{i\in\mathcal{S}_3} \left\{\left|Z_i - Z_{[2]}\right|\right\}$ because:
    1) For $Z_{[1]} \geq 0$, from Lemma 5, we get $Z_{[2]} \leq Z_{[i]} \leq Z_{[1]}$ for $i \in \{3,4,\ldots,N\}$. Also, $\left|Z_{[1]}\right| \geq \left|Z_{[2]}\right|$ implies $Z_{[2]} \geq -\left|Z_{[1]}\right| = -Z_{[1]}$. Hence, $-Z_{[1]} \leq Z_{[2]} \leq Z_{[i]}$ for $i \in \{3,4,\ldots,N\}$. Therefore, the node with the largest QI difference from $-Z_{[1]}$ is same as the node with the largest QI difference from $Z_{[2]}$. Hence, $[3] = \arg\max_{i\in\mathcal{S}_3} \{\mu_i'(3)\} = \arg\max_{i\in\mathcal{S}_3} \{\mu_i(3)\}$.
    2) Similarly, we can deduce the same for $Z_{[1]} < 0$.
- *Step* $k \geq 4$*:* The logic is similar and skipped.

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